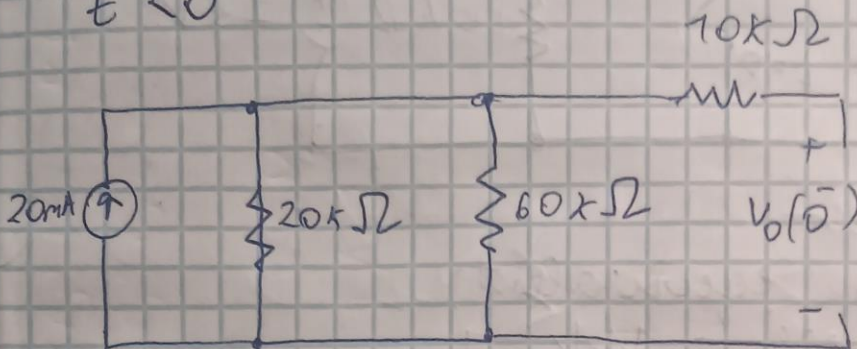


7.22

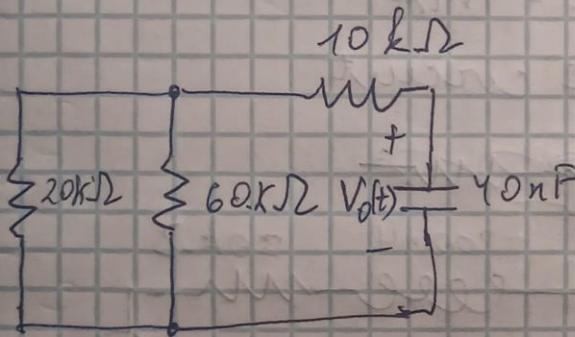
$t < 0$



$$V_0 = (20.000 \parallel 60.000) (20 \cdot 10^{-3}) =$$

$$= \frac{20 \cdot 10^{-3}}{\frac{1}{20.000} + \frac{1}{60.000}} = 300 \text{ V}$$

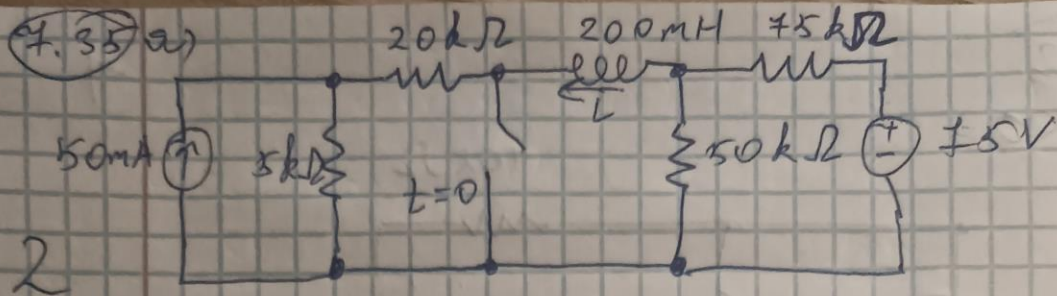
$t \geq 0$



$$R_{eq} = 10.000 + (20.000^{-1} + 60.000^{-1})^{-1} = 10.000 + 15.000 = 25.000 \text{ k}\Omega$$

$$\tau = (25 \text{ k}\Omega \cdot 40 \cdot 10^{-9} \text{ F}) = 1 \text{ ms}$$

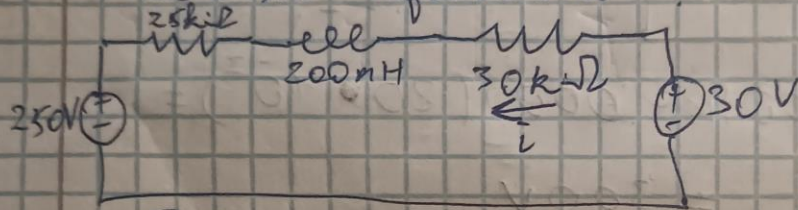
$$V(t) = 300 e^{-1000 t} \text{ V} \quad t \geq 0$$



2

~~for~~  $t < 0$

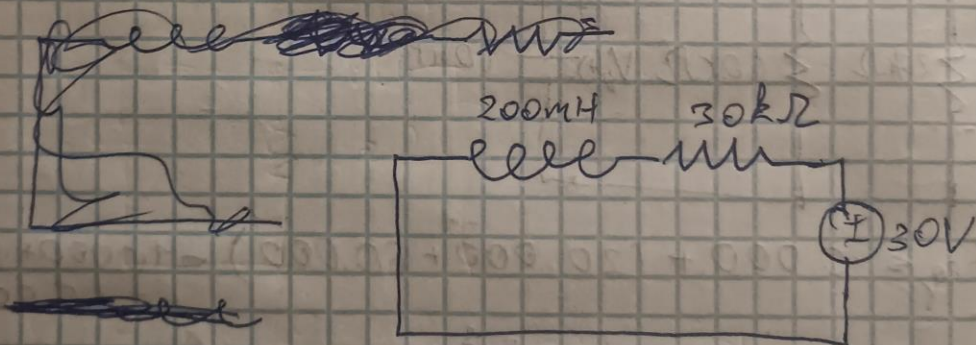
Thevenin equivalent:



$$\cancel{i(0^-)} \quad i(0^-) = \frac{30 - 250}{25k - 30k} = -4 \text{ mA}$$

$$i(0^-) = i(0^+) = -4 \text{ mA}$$

b)  $t > 0$  reduce circuit



$$i(\infty) = \frac{30}{30000} = 1 \text{ mA}$$

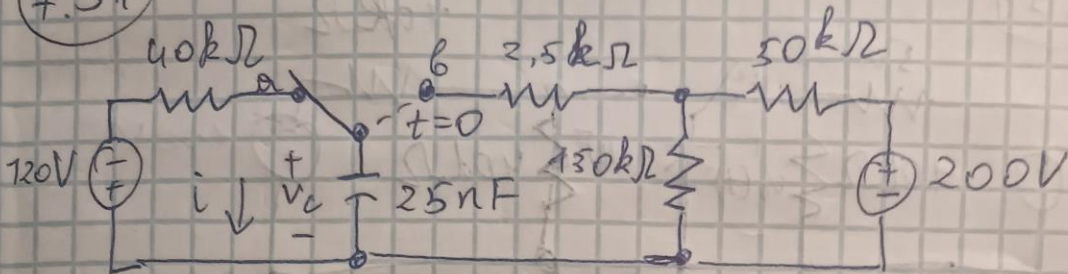
$$c) \tau = \frac{200 \cdot 10^{-3}}{30000} = 6.67 \mu\text{s}$$

$$d) i(t) = i(\infty) + [i(0^+) - i(\infty)] e^{-t/\tau} =$$



$$= 0.0001 + (-0.0001 - 0.0001)e^{-150000t} = 1 - 5e^{-150000t} \text{ mA}$$

7.51



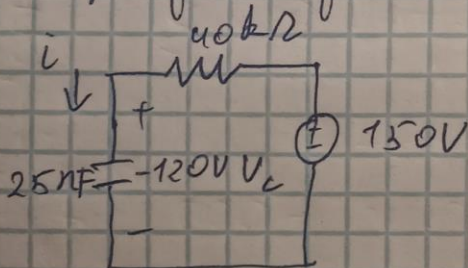
a)  $V_c(0^+) = -120 \text{ V}$

b)  $V_c(\infty) = \frac{150 \cdot 10^3}{200 \cdot 10^3} \cdot 200 = 150 \text{ V}$

c)  $V_{th} = 150 \text{ V}, R_{th} = 2500 + 150k \parallel 50k = 40k\Omega$

$\tau = 40.000 \cdot 25 \cdot 10^{-9} = 1 \text{ ms}$

simplified for  $t > 0$

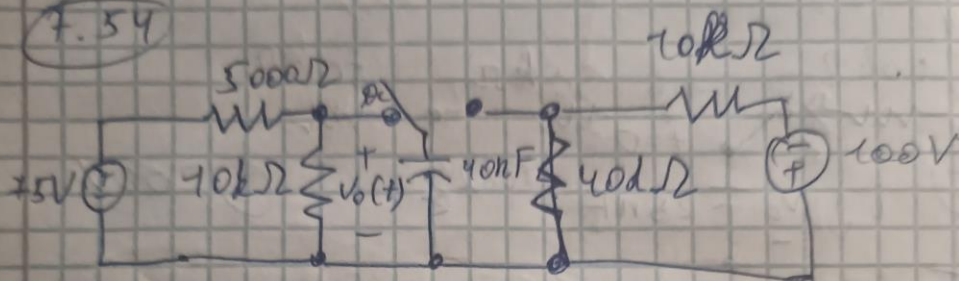


d)  $i(0^+) = \frac{150 + 120}{40 \cdot 10^3} = 6.75 \cdot 10^{-3} \text{ A}$

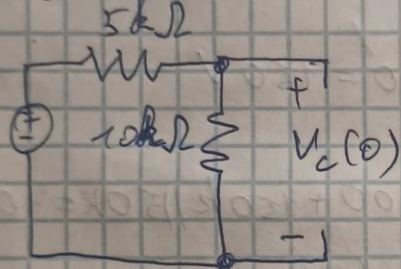
e)  $V_c = 150 + (-120 - 150)e^{-t/\tau} = 150 - 240e^{-10000t} \text{ V}, t \geq 0$

$$F) i = 25 \cdot 10^{-3} \cdot (-1000) \cdot (-2400 e^{-1000t}) = 6,75 e^{-1000t} \text{ mA} \quad t \geq 0^+$$

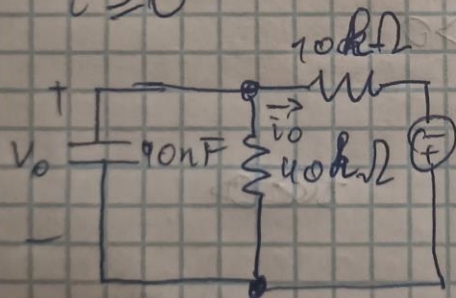
4) (7.54)



a)  $t < 0$



$t \geq 0$



$$V_o(\infty) = \frac{40.000}{50.000} \cdot (-100) = -80V$$

$$R_{eq} = \frac{1}{\frac{1}{40k} + \frac{1}{10k}} = 8k\Omega$$

$$\tau = R_{eq} C = 8000 \cdot 40 \cdot 10^{-9} = 3,2 \cdot 10^{-4} s$$



$$v_o(t) = v_o(\infty) + (v_o(0) - v_o(\infty))e^{-t/\tau} = -80 + (50 + 80)e^{-3125t} = -80 + 130e^{-3125t} \text{ V}$$

b)  $t \geq 0$

$$i_o = \frac{130e^{-3125t} - 80 + 100}{\cancel{10} \cdot 10^4} = 13e^{-3125t} + 2 \cdot 10^{-3} \text{ A}$$

5